

Kong AI Gateway — Kubernetes Deployment

Multi-environment topology (View 0) • Single-environment topology (View 1) • Plugin request path (View 2) — DXC architecture team review

Updated: 2026-05-13

Original: 2026-05-11

Gateway owner: Jesse Swensen

Gateway engineering: Axle

Source: 2026-05-11-kong-ai-gateway-k8s-deployment-diagram.md

Companion: 2026-05-11-dxc-arch-team-ask-kong-ai-gateway.md

2026-05-13 — **NEW** View 0 (multi-environment topology: Dev / Test / Prod) added following Jesse's decision to lock in a standard Dev → Test → Prod tier across all phases. Views 1 and 2 now describe the in-cluster shape *for a single environment*; that shape repeats per env. Konnect-native pattern: one Konnect org, one control plane per environment. Status: **⌘ OPEN — ARCH TEAM CONFIRMS, ROW 3** **2026-05-11** — Initial draft (View 1 topology, View 2 request path).

VIEW 0

Multi-Environment Topology — Dev / Test / Prod

NEW 2026-05-13

One Konnect organization carries three control planes — one per environment. Three separate K8s scopes attach as data planes, each controlled through the IaC repo under PR-gated promotion. View 1 (below) shows the in-cluster shape that repeats inside each of these three K8s scopes.

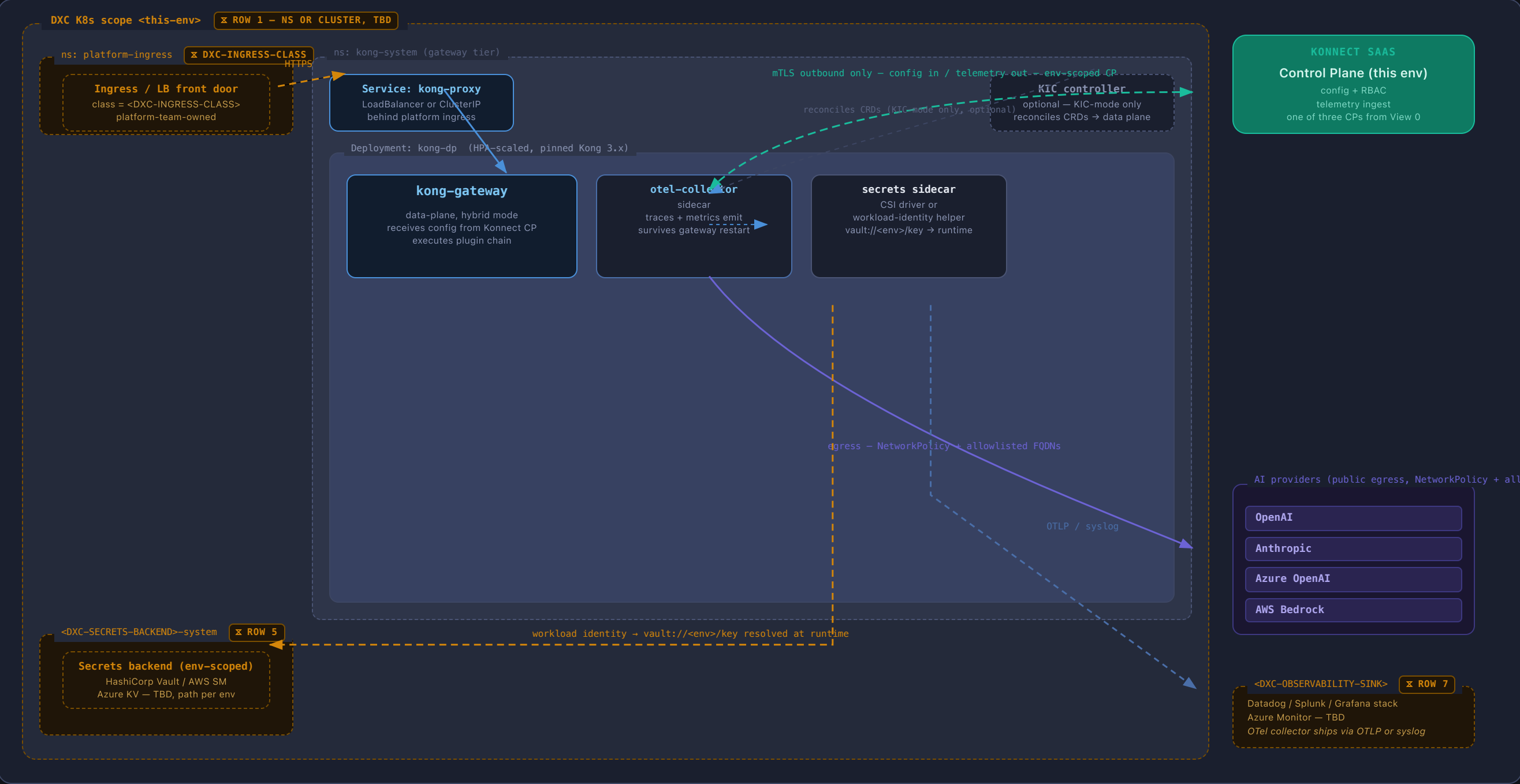


- **Single Konnect org, three control planes.** Modern Kong pattern replacing "Runtime Groups." Single billing, single audit trail, environments operationally isolated — a Dev config error cannot reach Prod data planes. **⌘ OPEN — ARCH TEAM ROW 3**
- **Secrets are scoped per env.** vault://dev/, vault://test/, vault://prod/ — no cross-env path collisions. Dev uses sandbox credentials; Prod uses production keys with strict rotation cadence.
- **IaC repo is the only path to any env.** No hand-edits in the Konnect UI in any environment. Prod is the hard rule; Dev / Test are operating discipline. Emergency hot-fixes get a 24h back-fill PR.
- **K8s scope per env is the platform team's choice.** Two shapes: (a) separate namespaces on a shared cluster; (b) separate clusters per env. Option (b) is the stronger isolation story for Prod at Tier 2+. **⌘ OPEN — ARCH TEAM ROW 1**
- **Phase transitions are config-only, not infrastructure re-architecture.** Phase 1 (observe) → Phase 2 (enforce) is a deck YAML delta promoted Dev → Test → Prod. No re-deploy. This is the property that makes the four-phase plan operationally cheap.
- **Network segmentation between envs is non-negotiable.** Prod data planes never reach Dev / Test services or vice versa. Consumed from the DXC platform team's network-segmentation story — not invented by the gateway team.

VIEW 1

Single-Environment Topology — In-Cluster Shape (Repeats per Env)

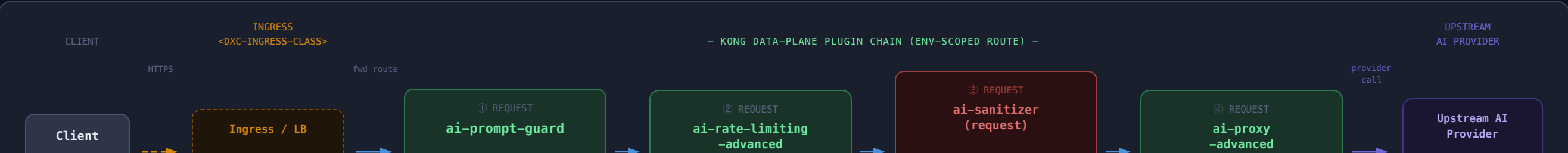
The in-cluster shape for *any one* of the three environments shown in View 0. Structure is identical across Dev / Test / Prod; what changes is the control-plane URL, provider-key vault scope, observability sink namespace, and scale (Prod goes multi-region at Tier 2+). Placeholders marked **⌘ DECISION PENDING** are open architecture decisions (Rows 1, 5, 7 of the ask doc).

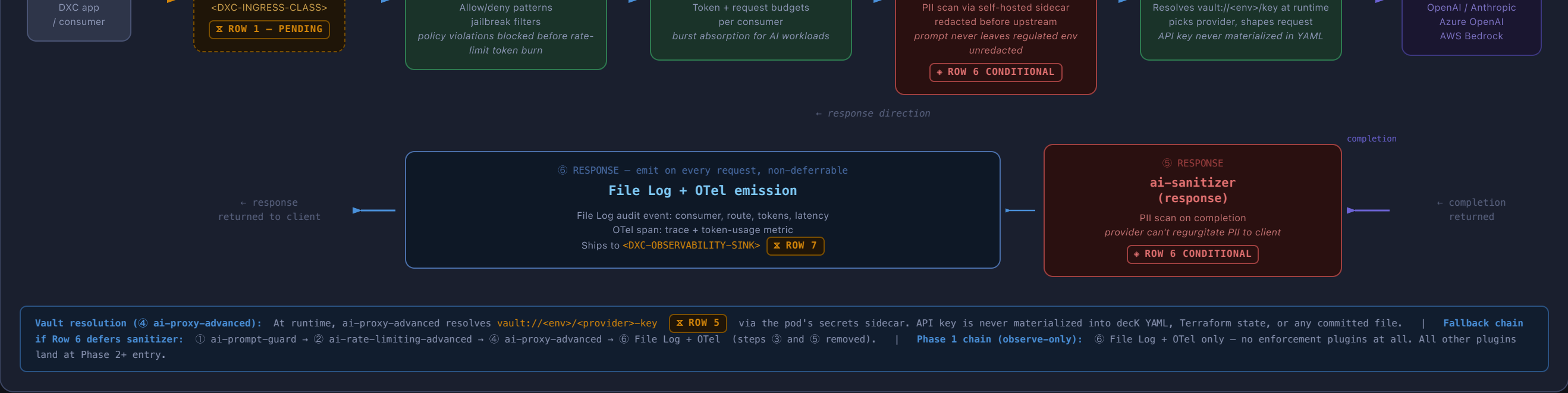


VIEW 2

Plugin Request Path — Recommended v1 Plugin Chain (Row 4)

Environment-agnostic: describes how a request traverses the plugin chain regardless of which env the route is in. **Phase determines what plugins are on the chain** (Phase 1 = File Log + OTel only; Phase 2 adds enforcement plugins); **env determines which CP and which vault-scoped keys are in play**. Order is load-bearing — policy violations do not consume token budgets; PII is redacted before leaving the cluster. Steps marked **⬢ ROW 6 CONDITIONAL** drop if compliance defers PII redaction past v1.





LEGEND

- Placeholder — decision pending (arch-team input required)
- Row 6 conditional — drops if compliance defers PII redaction
- Konnect SaaS control plane (one per env in View 0)
- Dev environment scope
- Test environment scope
- Prod environment scope
- Plugin (Recommended v1 set, pinned 3.x minor)
- mTLS outbound-only (Konnect CP → data plane / deck sync path)

Kong AI Gateway — K8s deployment shape. View 0: multi-env topology (Dev/Test/Prod). View 1: single-env in-cluster shape (repeats per env). View 2: plugin request path (env-agnostic). Every placeholder is an open decision; see the companion ask doc for options and tradeoffs.

GL-003 fallback mode — design system (GL-003-design-system.md) was unpopulated at render time (2026-05-13). Palette, type, and spacing are professional neutral technical defaults. Re-render once Iris populates GL-003 \$2 \$3 \$4.

Updated: 2026-05-13 · Original: 2026-05-11
Owner: Jesse Swensen (gateway owner) · Axle (gateway engineering)
Source diagram: 2026-05-11-kong-ai-gateway-k8s-deployment-diagram.md
Companion ask doc: 2026-05-11-dxc-arch-team-ask-kong-ai-gateway.md